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HACKATHON 2026

Idea Submission Template / Code. Create. Innovate.

Team Name	Team Raksha
Problem Statement No.	2
Problem Statement Title	Improving Safety at Large Public Events
Team Leader Name	Shounak Joshi

IDEA TITLE

Know the crowd. Catch the bottleneck. Stop the surge.

Cameras measure how dense each zone is, and how fast it moves.
It warns guards before a crush builds.

18

tracked at once

<1s

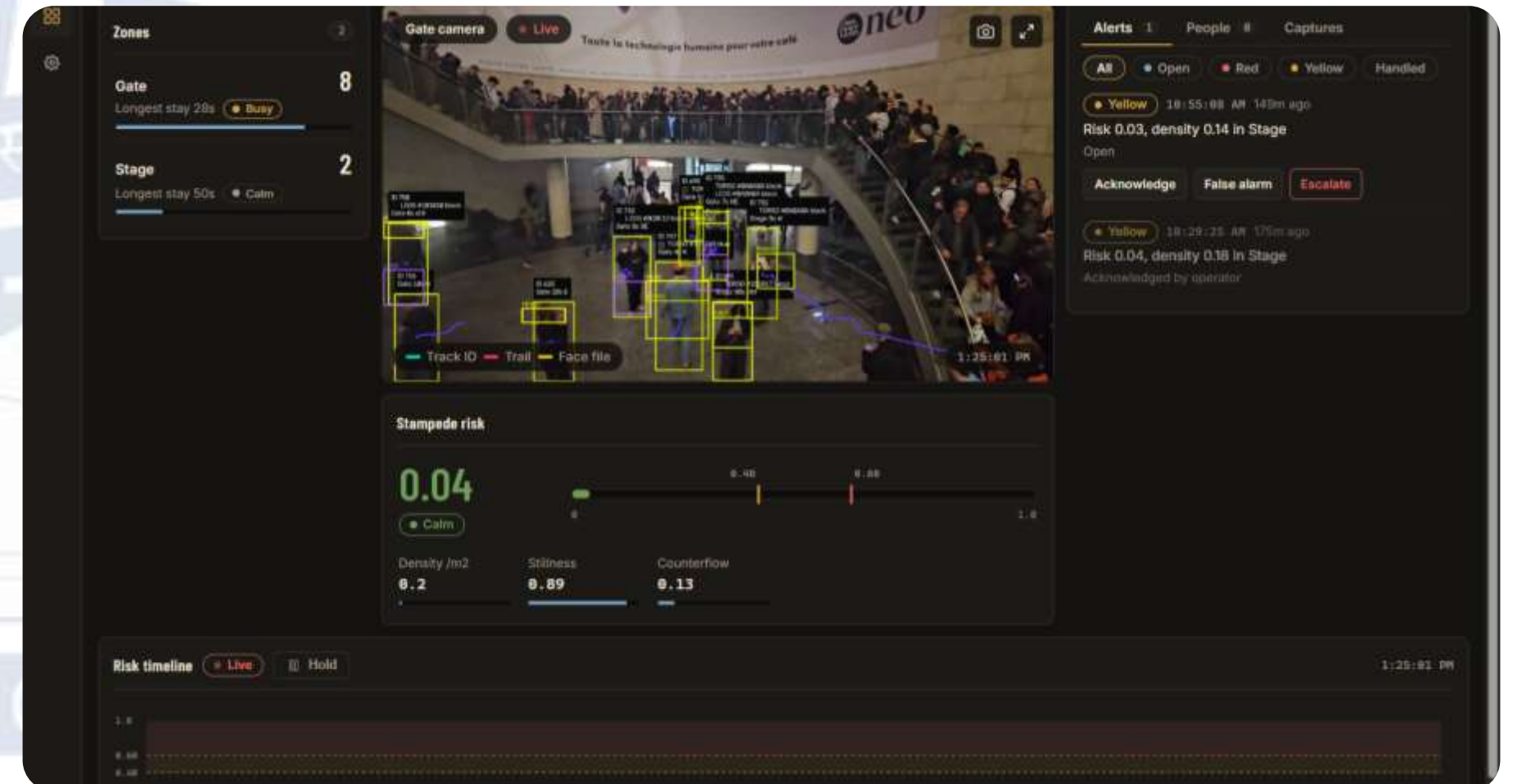
guard warned

15

video checks every
second

Watch it work

- 1 Live feed: every person counted, zone density tracked.
- 2 A zone packs up and stops: bottleneck flagged.
- 3 Guard warned in seconds; face file opens on demand.



The live wall, running on real station footage.

What the colours mean

● **Calm**
All is well.

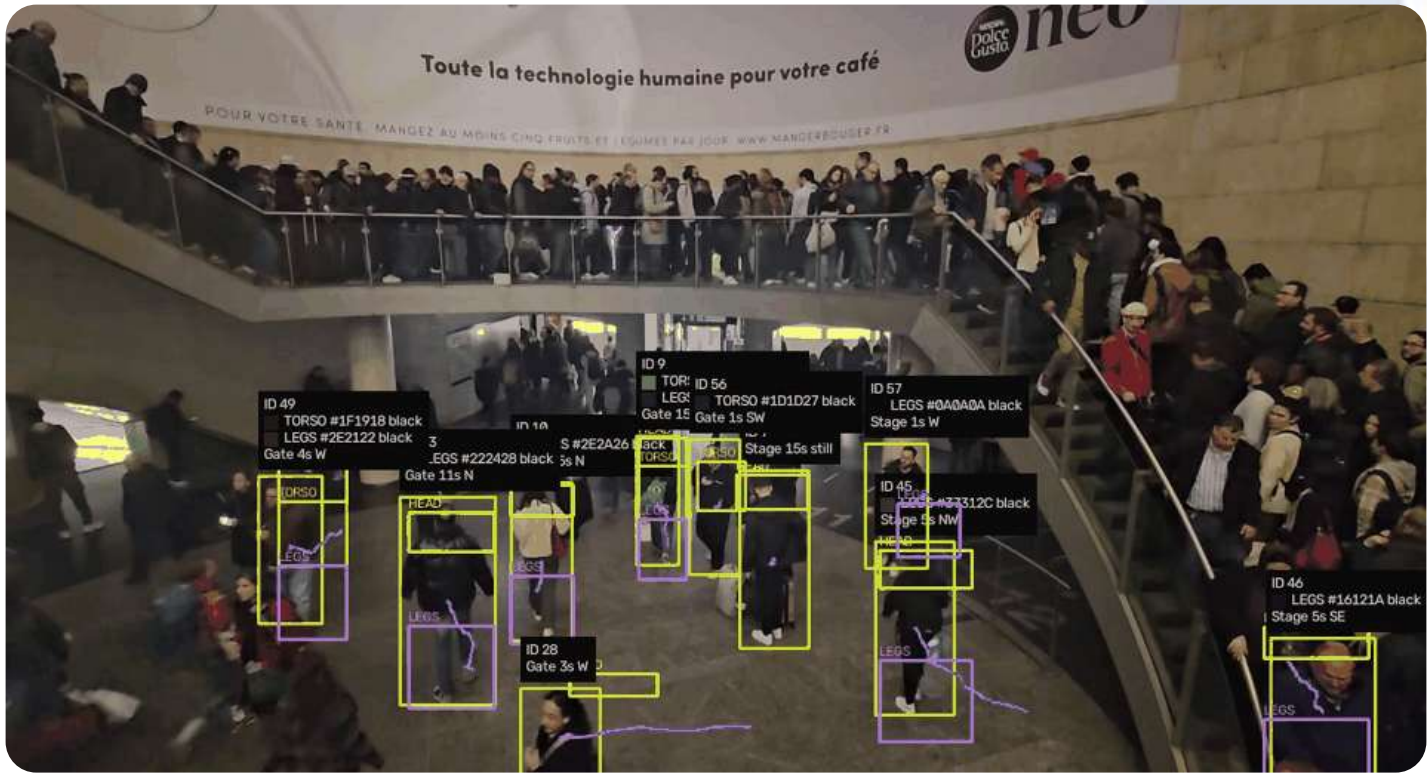
● **Busy**
Getting crowded. Guards get
an early warning.

● **Danger**
Crowd packed and stuck.
Clear the area now.

TECHNICAL APPROACH



In plain words: the camera watches, the software flags where the crowd stops moving, and it warns guards before a crush builds.



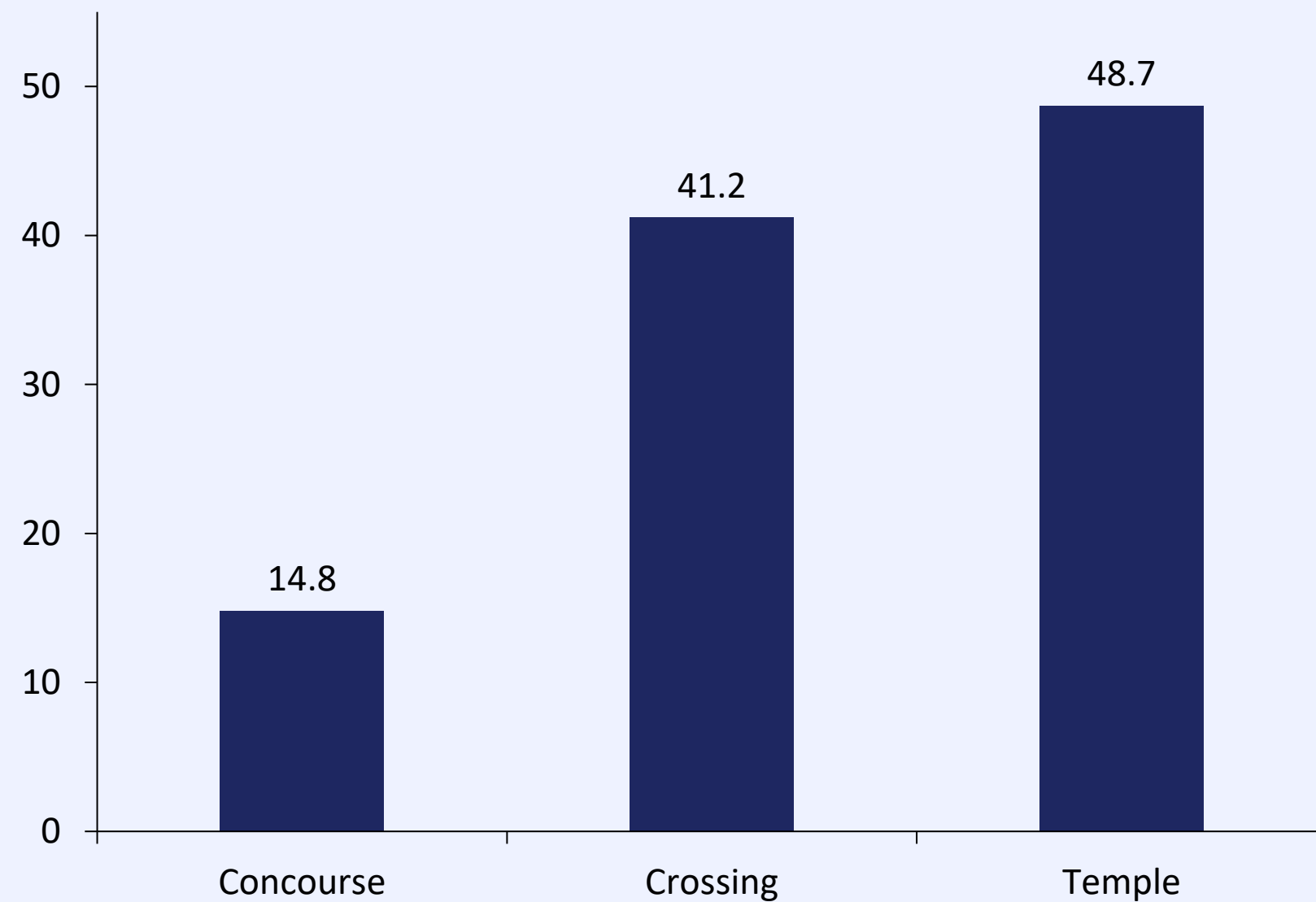
Keeps only good photos
Only sharp, front-facing photos are saved. Blurry ones are ignored, so matches stay reliable.

Tested on real video
Followed 18 people at once and 850 sightings across 2 minutes of station video.

Works with normal cameras
Any ordinary camera feed can be plugged in. No special equipment needed.

FEASIBILITY AND VIABILITY

Video checks per second (higher is better)



Tested on real crowd video from three places, all on one gaming laptop. Higher means smoother.

Instant

A face is matched in just 0.06 seconds.

Red alert

Fires when a zone packs tight and stops. The wall counts down the seconds to critical.

Runs on one laptop

Even the heaviest crowd runs smoothly.



IMPACT AND BENEFITS

When the alarm goes up

Yellow watch

Guards keep eyes on the zone. Crowd flow is slowed.

Red alert

Exits open, crowd diverted, announcement made.

After

A sealed evidence pack goes to the police.

Pilot targets: photo search in seconds, guard warned in under 1 second, every step logged.

Before

50 screens, one tired guard, and risks get missed.

After

One screen ranks every zone by risk, so guards only look where it matters.

Organizers

Spot a growing jam early, and tell guards which zone to clear.

Police

Find a lost child or a suspect from a photo in seconds. Full history in one click.

Public

Faces stay blurred in saved video.

RESEARCH AND REFERENCES

The science behind it

Finding and following people

Finds every person and follows each one. Uses YOLOv8 and ByteTrack, rated in 2025-26 studies as the best mix of speed and accuracy.

Code: github.com/ultralytics/ultralytics, github.com/ifzhang/ByteTrack

Matching faces

Compares a face with saved photos. Uses ArcFace and Faiss, which score 99.5 percent on the standard face test.

Code: github.com/deepinsight/insightface, github.com/facebookresearch/faiss

Predicting crushes

Crowd science models (AURORA and FTLE) spot danger early. A Delhi trial cut response time by 37 percent.

Ideas: AURORA warning model, FTLE bottlenecks, STAR-Crowd Delhi trial

Why India needs it

Hathras, Kumbh and railway station crushes all came from packed halls and blocked exits.

Test video: [Wikimedia Commons](#) public footage (station, crossing, temple)

Privacy built in

Blurred video, sealed proof

Saved video is face-blurred. Sharp stills stay sealed in the case file.

7-day routine wipe

Daily footage wipes after 7 days. Flagged incidents are exported first and kept for the police.

Tamper-evident log

Every action is hash-chained. Any edit leaves a trace.

No ID linking

Only people who choose to opt in can be matched.